

Requirements:

- Students must design a class or multiple classes for their assigned topic. Before writing code, students must analyze real-world scenarios (through observation, surveys, or domain research) to ensure their class design solves a practical problem and accurately reflects real-life entities.
- Students must implement as many fundamental concepts of Classes as possible such as:
 - Private/public + attributes/methods
 - Constructor, Overloading constructor, Destructor
 - Array of objects
 - Static attributes/member function
 - Operator overloading
 - This pointer
 - Class in Class
 - Friend of Classes
 - Object conversion
 - Memberwise assignment
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Output:

- UML diagram for each class
- Classnames.h for class definition. Students are encouraged to design a system with multiple interacting classes.
- .cpp files: The actual logic and definition of the methods declared in your Classnames.h files
- main.cpp

Geometry	Circle
	Point 2D
	Point 3D
	Rectangle
	Triangle
	Line 2D
	Line 3D
Mathematics	Fraction
	Complex Number
	2x2 Matrix
	FeetInches

Time	Date
	Time
	Duration
	Calendar
Business	Retail Item
	Inventory
	Shopping Cart
	Invoice
	Employee
Education	Students
	Lecturer
	Course
	Classroom
	Transcript
	Book
Booking	Flight Ticket
	Movie Ticket
	Hotel Room
Vehicle	Car
	Train
	Plane